Unbiased Estimator

$p^{true} = 0.6$

unknown

e.g. $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$

obs data $0, 1, 0, \ldots, 0$

$x_1, x_2, \ldots, x_n$

$\in$

$\{0,1\}$

estimate $\hat{p} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{n}(0+1+0+\cdots+0)$

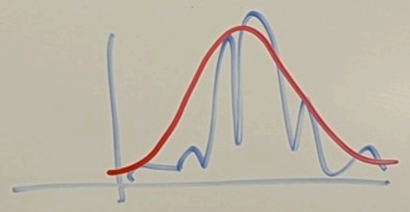
$= 0.55$

unbiased $E(\hat{p}) = p^{true}$ for all p^{true}

heights

e.g. $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ ← $Var(X) = E(X-\mu)^2$

↑ ↑ unknown parameters



unbiased →

$\hat{\mu} = \bar{x} = \frac{x_1 + \ldots + x_n}{n}$ ← 175 cm

$n-1$

$x_1 - \bar{x}, \; x_2 - \bar{x}, \ldots, x_n - \bar{x}$

$\sum_{i=1}^{n} (x_i - \bar{x}) = 0$

biased →

$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$ ← 20 cm

$<$ σ^2 on avg

↑ 3.6 on avg

unbiased

$\hat{\sigma}^2 = E\left(\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2\right) = \sigma^2$ on avg

adjusted

↑ lose 1 d.f.

Unbiased Estimator

true $= 0.6$ → θ

unknown

e.g. $X_1, \ldots, X_n \sim \text{Bernoulli}(\theta)$

$N(\mu, \sigma^2)$

$\vec{\theta} = (\mu, \sigma^2)$

obs data $0, 1, 0, \ldots, 0$

$x_1, x_2, \ldots, x_n$

$\in$

$\{0, 1\}$

estimate $\hat{p} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{n}(0+1+0+\cdots+0)$

$= 0.55$

unbiased $E(\hat{p}) = p^{true}$ for all p^{true}

$$P(\text{head}) = P_\theta(X_1 = 1) = \theta$$

(P_θ: $\theta = 0.7$; θ: $\theta = 0.7$)

e.g. $X_1, X_2, X_3 \overset{iid}{\sim} \text{Bernoulli}(\theta)$ ← indep, identical dist

obs data $x_1 = 1,\ x_2 = 0,\ x_3 = 1$

$$L(\theta) = P_\theta(\text{data}) = P(X_1 = x_1, X_2 = x_2, X_3 = x_3)$$

($x_1 = 1$, $x_2 = 0$, $x_3 = 1$)

likelihood function

$$= \underbrace{P(X_1 = x_1)}_{0 < \ldots < 1} P(X_2 = x_2) P(X_3 = x_3)$$

$$= \theta^{x_1}(1-\theta)^{1-x_1}\, \theta^{x_2}(1-\theta)^{1-x_2}\, \theta^{x_3}(1-\theta)^{1-x_3}$$

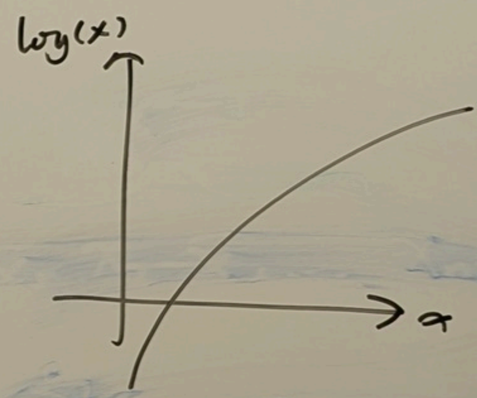
find $\theta^*_{MLE} = \max_{0<\theta<1} L(\theta) = \frac{2}{3}$

θ^*_{MLE}: can be biased

"$x_i \log(p) + (1-x_i)\log(1-p)$" cross-entropy loss

log-loss

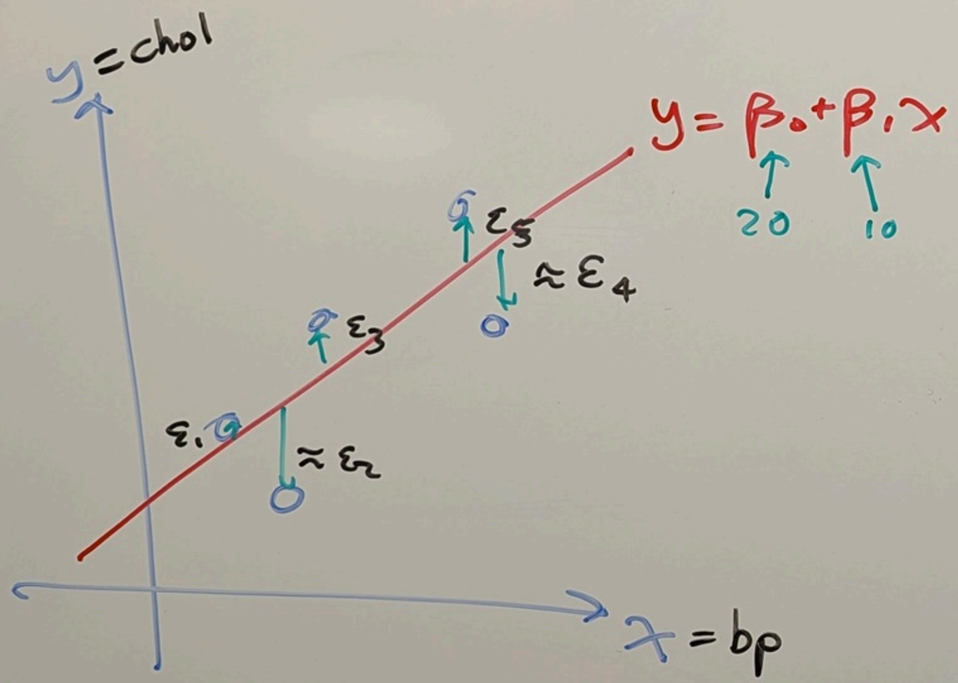
$$\log(ab) = \log(a) + \log(b)$$



e.g. $x_1 = x_2 = x_3 = 0$

$$\hat{p} = \frac{0}{3} = 0$$

Linear Reg Stat



$\vec{\beta}^*_{LS} \leftarrow$ min. $MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

Fitted model: $(chol) = 20 + 10(bp)$

70 ↑ 5

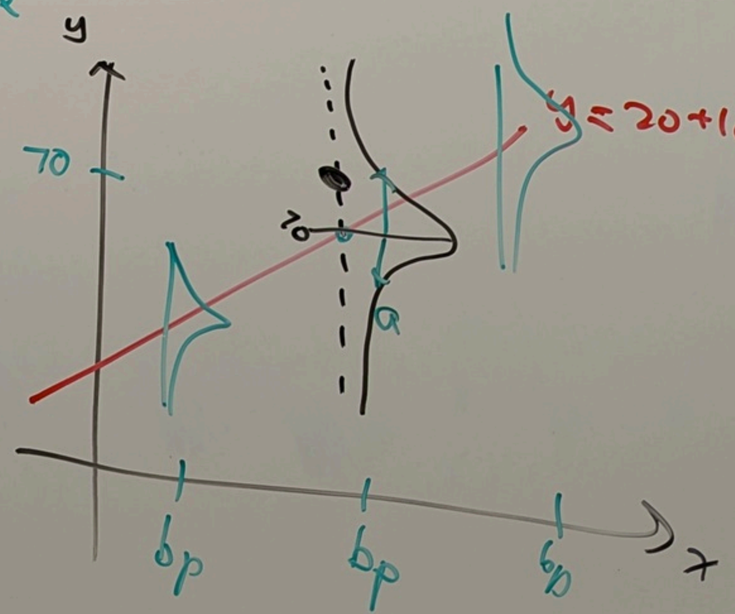
i^{th} person $chol = \underbrace{20 + 10(bp)}_{\text{avg pred. for bp=5}} + \underbrace{\varepsilon}_{\text{noise}}$

$= 70 + \varepsilon$

$Y_i^{chol} = \beta_0 + \beta_1 x^{bp} + \varepsilon$

$\varepsilon \sim N(0, \sigma^2)$

random variable



fixed random

data $(\vec{x}_1, y_1), \ldots, (\vec{x}_n, y_n)$

e.g. $X \sim N(0, \sigma^2)$

$5 + X \sim N(5, \sigma^2)$

$$Y_1 = \beta_0 + \beta_1 x_{11} + \ldots + \beta_d x_{1d} + \varepsilon_1 \sim N(\beta_0 + \beta_1 x_{11} + \ldots + \beta_d x_{1d}, \sigma^2)$$

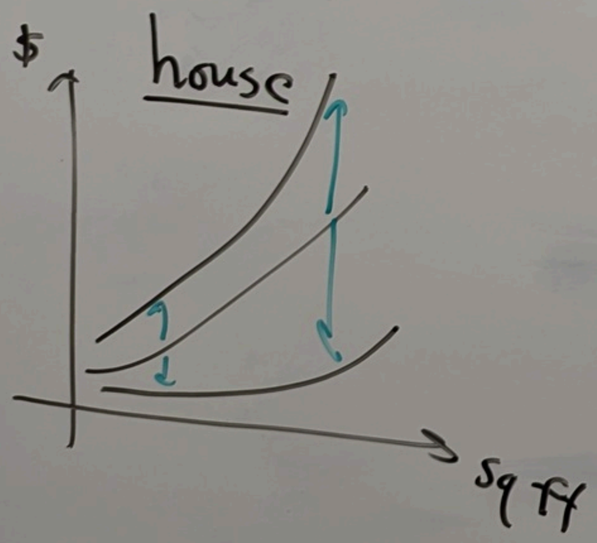
predictors

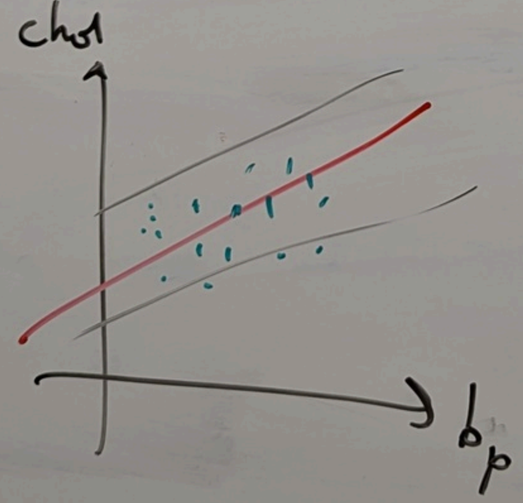
avg level

$\varepsilon_1 \sim N(0, \sigma^2)$ noise

$$Y_n = \beta_0 + \ldots + \beta_d x_d + \varepsilon_n$$

Bank 7:36 pm





$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

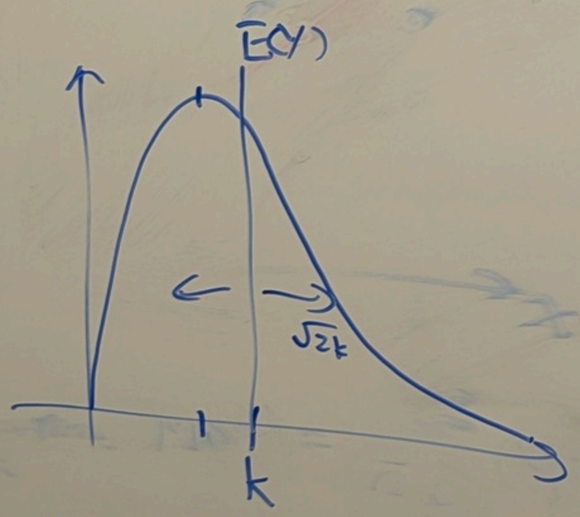
r_i or e_i

unbiased? biased?

χ^2

$$X_1, \ldots, X_k \sim N(0,1)$$

$$Y = X_1^2 + \ldots + X_k^2 \geq 0 \sim \chi^2_{df=k}$$



$E(Y) = k$

$Var(Y) = 2k$

Lin Reg

1+d=p parameters

$$Y_i = \beta_0 + \beta_1 x_1 + \ldots + \beta_d x_d + \varepsilon$$

$\varepsilon \sim N(0, \sigma^2)$

$$N(\beta_0 + \ldots, \sigma^2)$$

$$\hat{\sigma}^2_{MLE} = \frac{1}{n} \underbrace{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}_{SS_{error}}$$

$$E\left(\frac{\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2}{\sigma^2_{true}} \right) \sim \chi^2_{df = n-p}$$

1+d = # β's

$\frac{1}{n-p}$

in there, 1+d β's

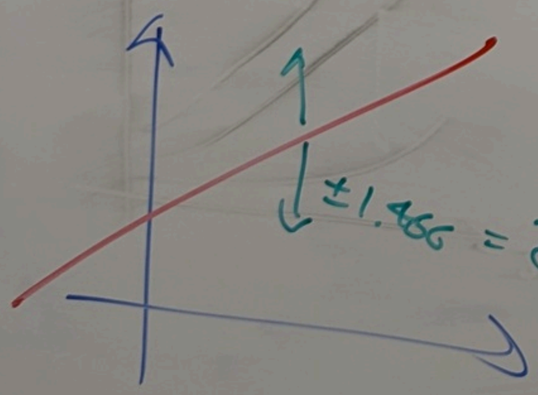
$$\hat{\sigma}^2_{unbiased} = \frac{1}{n-p} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$p = 1+d = \#$'s of β's $= \text{rank}(X)$

e.g. $Y_1, \ldots, Y_n \sim N(\underset{\beta_0}{\mu}, \sigma^2)$

unknown

$$\hat{\sigma}^2_{unbiased} = \frac{1}{n-1} \sum_{i=1}^{n} (y_i - \bar{y})^2$$



R: lm(y ~ 1)

$$Y_i = \beta_0 + \varepsilon_i \quad \varepsilon_i \sim N(0, \sigma^2)$$

$$\hat{\beta}_{MLE} = \bar{y}$$

$$\hat{\sigma}^2_{unbiased} = \frac{1}{n-p} \sum_{i=1}^{n} (y_i - \bar{y})^2$$

$p = 1$

Hypothesis Test

e.g. $X_1, \ldots, X_{100} \sim \text{Bernoulli}(\theta)$ ← unknown

null $H_0: \theta = 0.4$? (simple)

alt. $H_1: \theta \neq 0.4$

① test "statistic"

↑ a number you compute from data

$T = (X_1 + \ldots + X_{100}) = 31$

② need reference / benck mark distribution

31 vs 0.4?

without knowing θ

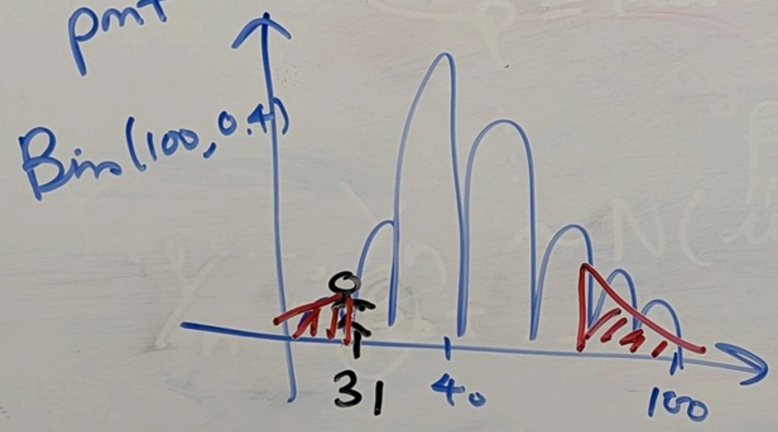
"Under H_0"

③ If H_0 is true

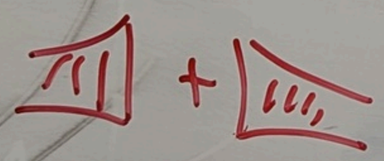
$$X_1, \dots, X_{100} \sim \text{Bern}(0.4)$$

$$T = (X_1 + \dots + X_{100}) \sim \text{Bin}(100, 0.4)$$

$$= 3,$$



④ p-value



$$= 0.07 \text{ (low)}$$

2-tails area

(maybe) H_0 is false!

Model Selection

large model

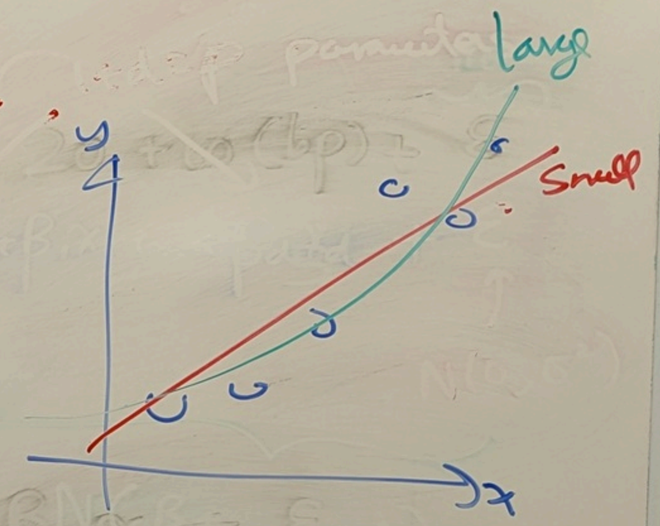
$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$

more flexible

(nested) smaller model

$y = \beta_0 + \beta_1 x + \varepsilon$

less flexible



$MSE_{large} < MSE_{small}$ -0.0001

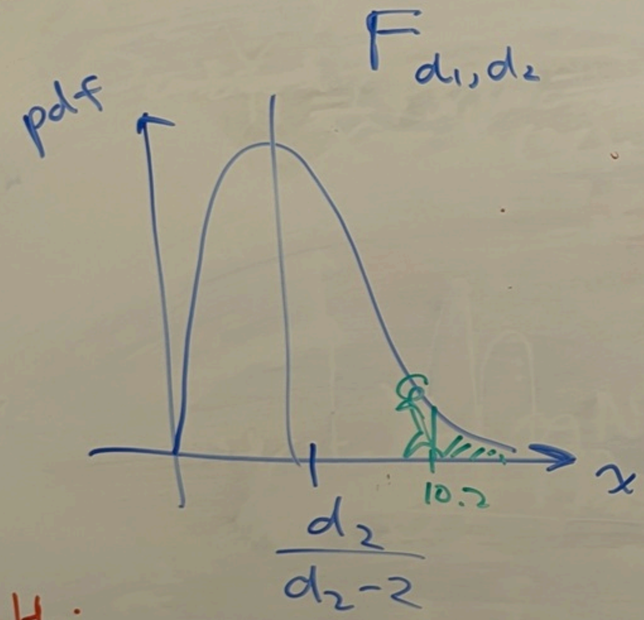
$R^2_{large} > R^2_{small}$

higher complexity

parameter 3 > 2 $+1$

model

F-Dist



$N(\mu, \sigma^2)$

H_0: small model is good enough, no need large model, $\beta_2 = 0$

p-value small $\leftrightarrow$ f is large

Reject H_0 $\leftrightarrow$ large model useful

H_1: otherwise

$p^{small} = 2$ $SS^{small} = 52$

$p^{large} = 3$ $SS^{large} = 32$

Under H_0:

$$f = \frac{\left(\frac{SS^{small} - SS^{large}}{p^{large} - p^{small}}\right)}{\left(\frac{SS^{large}}{n - p^{large}}\right)} \sim F_{d_1,d_2}$$

($d_1 \leftarrow$ numerator, $d_2 \leftarrow$ denominator)

f: test statistic

If large model much more useful

$$f \approx \frac{(large)}{(small)} = large$$

$f = 62.9$

not useful

$$f = \frac{(small)}{(large)} = small$$

